



SCI_LAUNCH_A2_W03 · LAUNCH Science

Autumn 2 Week 3 teacher note

Objective (verbatim): Stem-cell ethics discussion (links RE/PSHE).

Scheme of work: LAUNCH scheme of work, Autumn 2 week 3.

Pupil words: I am learning to weigh a possible benefit and risk of stem-cell research using evidence.

Reading gate: LAUNCH 11+; rough FK grade 7.4, age proxy ≈ 13.4 , 995 pupil-visible words excluding marked adult guidance. Formula is a rough screen, not an individual reading assessment.

Pearson foundation route: GCSE Biology 1BI0 Paper 1 and Combined Science 1SC0 Biology Paper 1, shared points 2.9 (medical stem-cell benefits and risks); consent is a research-ethics extension, not an extra spec point. Official specification links: <https://qualifications.pearson.com/content/dam/pdf/GCSE/Science/2016/Specification/gcse-biology-spec.pdf> and <https://qualifications.pearson.com/content/dam/pdf/GCSE/Science/2016/Specification/gcse-combinedscience-spec.pdf>. The workbook's teaching approach is not a new specification point. No Biology-only B/HT point is assumed for Combined Foundation.

Gatsby 4: An NHS research ethics coordinator considers participant safety, evidence and consent before a clinical study.

Sequence, success, retrieval

Arrival 5 · Starter 3 · I do 5 · We do 7 · We do 2 5 · Independent 8 · Review 4 · Exit 3 = 40 minutes. Three one-minute resets are inside We do, Independent and Review; Arrival and resets make about eight minutes for settling and transitions.

Success at Starter, checked at Exit: I can name a possible benefit and risk.; I can ask for evidence that tests a claim.; I can justify further research without claiming a cure.

Retrieval from SoW: "last term" uses the Autumn 1 block in the supplied 2026/27 workbook. If a literal previous academic term is required, staff must check actual teaching rather than invent it.

When	Question	Answer	Exact SoW cell
last lesson	What is the name for a cell gaining a specialised job?	Differentiation.	LAUNCH scheme of work, Autumn 2 week 2
2-4 weeks	What two cells follow mitosis and cell splitting?	Two matching daughter body cells.	LAUNCH scheme of work, Autumn 2 week 1
last term	In which direction do particles spread by diffusion?	From a higher to a lower concentration overall.	LAUNCH scheme of work, Autumn 1 week 4 (Aut1 W4)

Word meanings and examples (same in lesson and organiser):

Word	Tier	Definition	Example sentence
stem cell	3	a cell that can divide and form specialised cell types	Stem cells are being studied for possible medical uses.
benefit	2	a possible helpful result	A possible benefit would be replacing damaged cells.
risk	2	a possible harmful result or uncertain effect	A study must consider a risk to volunteers.



Word	Tier	Definition	Example sentence
evidence	2	information that tests or supports a claim	The trial needs evidence of both benefit and harm.

Faded example: I do full: A fictional proposal may replace damaged cells. A possible harm is unknown. A careful study must measure both benefit and risk; possible use is not proven treatment. We do part-done: Possible benefit: may replace damaged cells. Possible risk: . Evidence needed: a fair check of benefit and harm. Removed step: Identify an uncertain harmful result as a risk, without stating it has already happened. Independent uses no filled answer. Fade next: Next time leave the evidence-needed column blank so the pupil decides what test or observation would discriminate a claim.

Post-I-do show-me: Show me: is “it might help” evidence that the treatment already works? Expected: No. It is a possible benefit; evidence from an appropriate study is needed. Every pupil shows a board, point or card. Re-model the named step if insecure.

F/M/H and named stretch

We do F support: Give three colour-coded headings and a fictional benefit/risk card; pupil supplies the missing risk and one request for evidence.

We do M: Explain why a possible benefit is not evidence that a treatment works.

We do H stretch: Ask whether “support more research” and “approve a treatment for use” require the same strength of evidence; justify the distinction.

Independent F: Sort the fictional proposal statements under possible benefit, possible risk and evidence needed. Stem: A possible benefit is ...; a possible risk is ...

Independent M: Explain why a proposal needs evidence of both effect and harm before a decision. Stem: The claim is not yet proven because we need evidence that ...

Independent H: Evaluate whether to support further research with a qualified reason, risk and evidence request. Stem: I would support/review further research if ...; however, before treatment use I would need ...

Go deeper: If a careful new study showed no improvement, how should the panel change its recommendation and why?

Named early-finisher stretch — Research panel minute: Separate further research from routine treatment; explain the evidence needed for each. This is a harder correction/evaluation, not a repeated item.

On-screen interactions: Which panel statement is not based on evidence? Choose it, then state the better question. uses native selects with a check/reset and aria-live feedback; We do 2 is a native 4-option all-pupil hinge without auto-grading. No storage. Original labelled diagram is simplified, not to scale. Its labels are checked in the fact ledger.

SoW practical status: no core practical is claimed. Prepared model pieces are an optional handling rehearsal with the pupil-facing Risk/Safe lines below.

AfL hinge and misconceptions

Question at We do 2: A new fictional cell study reports improved movement but does not report harmful effects. Which review is justified?

Option	Correct?	Diagnosis
A. The benefit can establish safety despite the missing report on harms.	no	Treats an observed benefit as proof of safety without harm data.



Option	Correct?	Diagnosis
B. The missing report proves the study cannot have produced any benefit.	no	Treats missing harm reporting as proof that the reported benefit did not occur.
C. The improved movement proves the study will work for every person.	no	Generalises one study result to every person.
D. The benefit needs checking alongside missing risk evidence before a decision.	yes	Uses the new case accurately.

Decision: About 3 in 4 or more correct: proceed; otherwise re-model a fictional claim card beside an empty evidence column and ask the parallel hinge. No auto-grade. Parallel hinge: Another fictional study reports harms but no benefit measure. Can it establish a favourable benefit-risk balance?

Answer: no; both kinds of evidence are needed.

Misconception	How to spot	Correction
A reported benefit in one study proves a treatment will work for everyone.	Pupil chooses C, claiming benefit for every person.	Ask for further evidence of benefit across people and settings.
A reported benefit establishes safety without harm evidence.	Pupil chooses A, treating benefit as proof of safety without harms data.	Review benefits and missing risk evidence together before a decision.

Planned adaptations

Broad area	EEF five-a-day	Concrete action
Cognition & Learning	scaffolding	Scaffolding: Use benefit/risk/evidence columns with one completed model sentence, then hide one column cue.
Communication & Interaction	flexible grouping	Flexible grouping: Give a structured partner reply and a pass option; accept AAC, sign or spoken view of the fictional case.
SEMH	cognitive/metacognitive strategies	Cognitive/metacognitive strategies: No request for personal belief or family medical history; private check-in, quiet alternative and regulated return.
Sensory/Physical	technology	Technology: Offer large text or immersive-reader-friendly fictional cards; pupil may speak or type the reason.
EAL (language, not SEN)	explicit instruction	Explicit instruction: Define benefit, risk and evidence with concrete examples; allow planning in any language. Keep M and H scientific demand available.

EAL language support remains available at M and H; the challenge stays open. Speak before write; the organiser's "My language" column is blank, without machine translation. Accept point, sign, speak, draw, type, write, exact scribing and staff-captured photo. Offer speech-to-text, TTS and immersive reader.

TA Focus and prompt ladder



At Independent choose at lesson start: teacher works with supported pupils and TA holds group, or TA supports and teacher holds group. Swap across the week. No public cold-calling is required; this is an invitation bank.

I do: "Which statement is a possible result, and which is measured evidence?"

We do: "The risk column is blank. What question could the panel ask?"

Independent: "What is your reason for more research or a pause? Which evidence would test it?"

Least prompt first; no physical prompting: Wait five seconds for the pupil to choose a column. → Point to a card heading and the last completed model sentence. → Ask "What would a careful panel need to find out?" → Model the first benefit card only; pupil places the risk card. → Place one card together, then invite a new evidence request independently.

Exit handoff: adult states deepest prompt level, valid response mode and what pupil did independently; other adult chooses the next prompt.

Route	Invitation	Expected answer
F	What word means a possible good result?	Benefit.
F	What word means information used to test a claim?	Evidence.
M	Why is "might help" not proof of a treatment?	It states a possibility; evidence of benefit and risk must be collected.
M	Name one information request for the fictional study.	Ask for measured benefit or safety outcomes and how they were checked.
H	How could new evidence change a panel decision?	It could increase or reduce confidence about benefit and risk, changing whether further work is supported.
H	Why can further research be supported without approving routine treatment?	Research tests uncertainty under review; routine use needs stronger evidence of safety and benefit.

Feedback, EHCP broad areas and regulation

Codes exactly: VF · WS · I · NS(+) · E · R · // · ?. VF at We do or Review; WS when supported with recorded prompt level; I when independent at Independent/Exit; NS(+) precise next step at Exit; // a real self-edit at Review; ? adult reads the changed words back on request; R after the pupil responds to NS(+) next lesson and staff judge, never automatic; E only for an actual curriculum observation.

What I record and where: note hinge response split and the named misconception on the staff quick-record grid; record actual code and prompt level in feedback. E: evidence captured on the school's evidence app. Never gather evidence during distress or store pupil data in HTML.

Pathways: high support, VF/WS regular. Build: WS/I mix. Grow: mostly independent, self-edit and response. Launch: independent self-assessment.

Regulation: Check in before the hinge, top rung and Independent. At the agreed pause signal give space and a single return step. Collect no evidence while a pupil is distressed. Re-enter at the last completed step, with one small choice and private correction.

lesson record poster: LAUNCH_SCI_A2_W03. Staff device or projector only; pupil phones not required.

Lundy Exit: pupil can say, point, sign, draw or write; teacher hears the answer and changes the next model/prompt. Four answers: goal at top, actual Review feedback, Exit NS(+), success criteria at



Starter and Exit.

Sources, checks and limits

Every scientific and diagram label claim is listed in LESSON_SOURCE.json with a named official source and status. False distractors are deliberately incorrect and diagnosed. Read source claims in context; no medical claim about an individual. The proposal and panel are invented classroom scenarios. No pupil has to state a personal belief, disclose health information or decide real treatment.

Prepared model handling · practical safety

Kit: Large prepared fictional benefit, risk and evidence cards, desk mat

Method: Sort the prepared fictional cards under the three headings.

Pupil Risk: Loose model cards could fall onto the floor.

Pupil Safe: Keep cards on the desk mat and pick up any that fall.

Staff confirm the local school risk assessment and exact kit; this does not assert a SoW core practical.

SLT planning and evaluation

Lesson code: SCI-LAUNCH-A2-W03

SoW reference: LAUNCH scheme of work, Autumn 2 week 3

Exact SoW outcome: Stem-cell ethics discussion (links RE/PSHE).

Evidence (scheme of work): Edexcel 1BI0 Paper 1 (1F) topic tests; AQA UAS science units. F: topic quizzes, practical write-ups. S: Topics 1 to 3 assessment (Dec).

Evidence in this lesson: Edexcel 1BI0 Paper 1 (1F) topic tests; AQA UAS science units. F: topic quizzes, practical write-ups. S: Topics 1 to 3 assessment (Dec).

Where it shows in books: Independent observation/result table or labelled explanation; review edits and exit response beside the original work.

Label evidence on the school's evidence app with this code: SCI-LAUNCH-A2-W03. For spoken, practical or photographic evidence, record the same code with the evidence.

Arrival (5 min): Look at this week's fictional stem-cell proposal card. Point to one detail you notice. Staff: Offer a private, predictable start. Do not require previous knowledge.

Starter (3 min): Recall three ideas from earlier lessons. Show one answer at a time. Staff: All pupils respond; reveal the answers only after attempts. Revisit success criteria.

I Do (5 min): Watch the full model. Point to the step that answers today's question. Staff: Model one complete example; check every pupil's response to the show-me question.

We Do (7 min): Complete the missing step in the part-done model. Speak, then match the cards. Staff: Wait five seconds, gesture, then cue if needed. Support the route without supplying the missing answer.

We Do 2 (5 min): Choose one hinge answer. Show it at the same time as the group. Staff: Collect simultaneous responses, apply the decision rule and parallel hinge in the staff note.

Independent (8 min): A fictional stem-cell proposal lacks safety results. Decide what to test next. Staff: At lesson start choose one adult with supported pupils and one holding the group; swap during the week. Record independence and prompt level.

Review (4 min): Tell a partner what you changed after feedback. Mark a real edit // and read it back when asked ?. Staff: Give a precise VF or NS(+) only when it actually happens. Invite a true self-edit



and read-back.

Exit (3 min): Show what you know with a model and explain one reason. Staff: Hear the answer privately in a valid mode and use it to change the next example or prompt.

Planned adaptation — cognition learning: Scaffolding: Use benefit/risk/evidence columns with one completed model sentence, then hide one column cue.

Planned adaptation — communication interaction: Flexible grouping: Give a structured partner reply and a pass option; accept AAC, sign or spoken view of the fictional case.

Planned adaptation — semh: Cognitive/metacognitive strategies: No request for personal belief or family medical history; private check-in, quiet alternative and regulated return.

Planned adaptation — sensory physical: Technology: Offer large text or immersive-reader-friendly fictional cards; pupil may speak or type the reason.

Planned adaptation — eal: Explicit instruction: Define benefit, risk and evidence with concrete examples; allow planning in any language. Keep M and H scientific demand available.

Planned adaptation — eef strategy: cognition learning: scaffolding; communication interaction: flexible grouping; semh: cognitive/metacognitive strategies; sensory physical: technology; eal: explicit instruction

Staff: confirm the planned adaptations actually used and record any changes below.

If behind — must teach core: A fictional stem-cell proposal lacks safety results. Decide what to test next. Preserve the individual evidence required above.

If behind — shorten or merge: After a secure hinge, shorten repeated oral rehearsal. If needed, merge the exit reflection into the Arrival of Autumn 2 W04. Keep the core evidence and any safety teaching; record the change and coverage impact.

If behind — next lesson: Autumn 2 W04: Describe DNA, genes & chromosomes.

If behind — dependency: Pass on today's checked understanding and unresolved hinge errors for the next starter. Do not replace incomplete individual work with a group product.

Staff evaluation — date taught:

Staff evaluation — delivered as planned: ☐ Yes ☐ Partly ☐ No

Staff evaluation — criteria met: ☐ All ☐ Most ☐ Some

Staff evaluation — adaptations made and why:

Staff evaluation — change from sow reason and coverage impact:

Staff evaluation — evidence banked on the school's evidence app:

Individual target opportunities

Cognition & learning: We do: Identify an uncertain harmful result as a risk, without stating it has already happened. Record the pupil's own sequencing, attention or independence target, including the prompt level needed.

Communication & interaction: Check: respond to "A new fictional cell study reports improved movement but does not report harmful effects. Which review is justified?" in the pupil's agreed mode. Evidence an individual communication/turn-taking target without requiring speech.

SEMH: Arrival: Look at this week's fictional stem-cell proposal card. Point to one detail you notice. Use the agreed predictable start or pause/re-entry plan; evidence the pupil's own regulation or engagement target privately.



Sensory/physical: Independent: A fictional stem-cell proposal lacks safety results. Decide what to test next. Offer the agreed accessible input, enlarged material or adapted equipment; record the pupil's target-related access/independence, not a diagnosis.

EAL: Model/shared rehearsal: use stem cell, benefit, risk, evidence with the actual example; allow first-language planning and oral rehearsal before the independent response. Record the agreed language target; EAL is not SEN.

Capture E against the pupil's own IEDP target on the school's evidence app; the IEDP carries their EHCP-linked and Boxall targets, so one record serves every plan. Follow the pupil's IBSP where it differs from these routines.